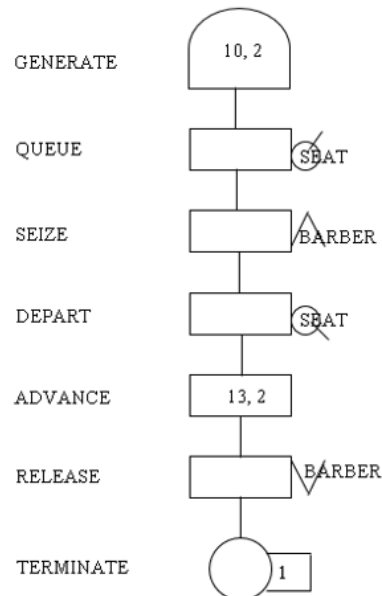


Lab No. 7

Create a GPSS model and program to simulate a barber shop for a day (9 a.m. to 4 p.m.), where a customer enters the shop every 10 ± 2 minutes and a barber takes 13 ± 2 minutes for a haircut.

Source Code:

```
GENERATE 10,2
QUEUE SEAT
SEIZE BARBER
DEPART SEAT
ADVANCE 13,2
RELEASE BARBER
TERMINATE 1
GENERATE 420
TERMINATE 1
START 1
```



Output:

GPSS World - [Untitled Model 1.1.1 - REPORT]

File Edit Search View Command Window Help

GPSS World Simulation Report - Untitled Model 1.1.1

Friday, June 19, 2026 18:41:55

START TIME	END TIME	BLOCKS	FACILITIES	STORAGES
0.000	25.893	9	1	0

NAME	VALUE
BARBER	10001.000
SEAT	10000.000

LABEL	LOC	BLOCK TYPE	ENTRY COUNT	CURRENT COUNT	RETRY
1	GENERATE	2	0	0	
2	QUEUE	2	0	0	
3	SEIZE	2	1	0	
4	DEPART	1	0	0	
5	ADVANCE	1	0	0	
6	RELEASE	1	0	0	
7	TERMINATE	1	0	0	
8	GENERATE	0	0	0	
9	TERMINATE	0	0	0	

FACILITY	ENTRIES	UTIL.	AVE. TIME AVAIL.	OWNER	PEND	INTER	RETRY	DELAY
BARBER	2	0.561	7.262	1	3	0	0	0

QUEUE	MAX CONT.	ENTRY ENTRY(0)	AVE. CONT.	AVE. TIME	AVE. (-0)	RETRY
SEAT	1	1 2	1	0.132	1.707	3.413 0

CEC	XN	PRI	M1	ASSEM	CURRENT	NEXT	PARAMETER	VALUE
3	0		22.480	3	3	4		

FEC	XN	PRI	BDT	ASSEM	CURRENT	NEXT	PARAMETER	VALUE
4	0		31.522	4	0	1		
2	0		420.000	2	0	8		

For Help, press F1 Report is Complete.

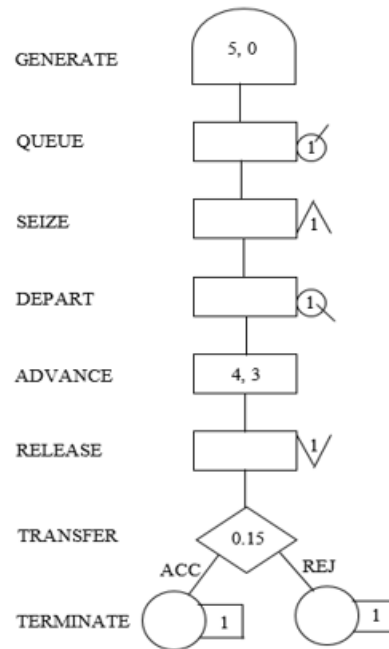
Lab No. 8

A machine tool in a manufacturing shop is turning out parts at the rate of every 5 minutes. When they are finished, the parts are sent to an inspector, who takes 4 ± 3 minutes to examine each one and rejects 15% of the parts. Draw and explain a block diagram for it and write a GPSS program to simulate using the concept of FACILITY.

Source Code:

```

GENERATE 5,0
QUEUE INSQ
SEIZE INSPECTOR
DEPART INSQ
ADVANCE 4,3
RELEASE
INSPECTOR
TRANSFER
0.15,REJ,ACC
ACC TERMINATE 1
REJ TERMINATE 1
    
```



Output:

GPSS World - [Untitled Model 1.1.1 - REPORT]

File Edit Search View Command Window Help

GPSS World Simulation Report - Untitled Model 1.1.1

Friday, June 19, 2026 18:47:20

START TIME	END TIME	BLOCKS	FACILITIES	STORAGES
0.000	11.286	9	1	0

NAME	VALUE
ACC	8.000
INSPECTOR	10001.000
INSQ	10000.000
REJ	9.000

LABEL	LOC	BLOCK TYPE	ENTRY	COUNT	CURRENT	COUNT	RETRY
	1	GENERATE	2		0	0	
	2	QUEUE	2		0	0	
	3	SEIZE	2		1	0	
	4	DEPART	1		0	0	
	5	ADVANCE	1		0	0	
	6	RELEASE	1		0	0	
	7	TRANSFER	1		0	0	
ACC	8	TERMINATE	0		0	0	
REJ	9	TERMINATE	1		0	0	

FACILITY	ENTRIES	UTIL.	AVE.	TIME	AVAIL.	OWNER	PEND	INTER	RETRY	DELAY
INSPECTOR	2	0.557	3.143	1	2	0	0	0	0	0

QUEUE	MAX	CONT.	ENTRY	ENTRY(0)	AVE.	CONT.	AVE.	TIME	AVE.	(-0)	RETRY
INSQ	1	1	2	1	0.114	0.643	1.286	0			

CEC	XN	PRI	M1	ASSEM	CURRENT	NEXT	PARAMETER	VALUE
	2	0	10.000	2	3	4		

FEC	XN	PRI	BDT	ASSEM	CURRENT	NEXT	PARAMETER	VALUE
	3	0	15.000	3	0	1		

Lab No. 9

A **GENERATE** block is used to represent the output of the machine by creating one transaction every five units of time. A **QUEUE** block places the transaction in the queue and **SEIZE** block allows a transaction to engage a facility if it is available. If more than one inspector is available, the transaction leaves the queue which is denoted by **DEPART** block and enters into **ADVANCE** block. An **ADVANCE** block with a mean of 4 and modifier of 3 is used to represent inspection. The time spent on inspection will therefore be any one of the values 1, 2, 3, 4, 5, 6 or 7, with equal probability given to each value. Upon completion of the inspection, **RELEASE** block allows a transaction to disengage the facility and transaction go to a **TRANSFER** block with a selection factor of 0.15, so that 85% of the parts go to the next location called **ACC**, to represent accepted parts and 15% go to another location called **REJ** to represent rejects. Both locations reached from the **TRANSFER** block are **TERMINATE** blocks.

Source Code:

```
GENERATE 5,0,,100  
QUEUE INSQ  
SEIZE INSPECTOR  
DEPART INSQ  
ADVANCE 4,3
```

```
RELEASE INSPECTOR  
TRANSFER 0.15,REJ,ACC
```

```
ACC TERMINATE 1  
REJ TERMINATE 1  
START 100
```

Output:

GPSS World - [Untitled Model 2.1.1 - REPORT]

File Edit Search View Command Window Help



GPSS World Simulation Report - Untitled Model 2.1.1

Friday, June 19, 2026 18:50:30

START TIME	END TIME	BLOCKS	FACILITIES	STORAGES
0.000	505.643	9	1	0

NAME	VALUE
ACC	8.000
INSPECTOR	10001.000
INSQ	10000.000
REJ	9.000

LABEL	LOC	BLOCK TYPE	ENTRY COUNT	CURRENT COUNT	RETRY
	1	GENERATE	101	0	0
	2	QUEUE	101	0	0
	3	SEIZE	101	1	0
	4	DEPART	100	0	0
	5	ADVANCE	100	0	0
	6	RELEASE	100	0	0
	7	TRANSFER	100	0	0
ACC	8	TERMINATE	17	0	0
REJ	9	TERMINATE	83	0	0

FACILITY	ENTRIES	UTIL.	AVE. TIME	AVAIL.	OWNER	PEND	INTER	RETRY	DELAY
INSPECTOR	101	0.826	4.133	1	101	0	0	0	0

QUEUE	MAX	CONT.	ENTRY	ENTRY(0)	AVE.CONT.	AVE.TIME	AVE.(-0)	RETRY
INSQ	2	1	101	42	0.206	1.029	1.762	0

CEC XN	PRI	M1	ASSEM	CURRENT	NEXT	PARAMETER	VALUE
101	100	505.000	101	3	4		

FEC XN	PRI	BDT	ASSEM	CURRENT	NEXT	PARAMETER	VALUE
102	100	510.000	102	0	1		

For Help, press F1

Report is Complete.

Lab No. 10

Parts are being made at the rate of one every 10 minutes. They are of two types; A and B. Mixed randomly with 10% being type B. Separate inspectors inspect each type. Inspection time for A is 6 ± 2 minutes and for B is 10 ± 2 minutes. Both inspectors reject 10% of parts. Simulate for 100 parts.

Source Code:

```
GENERATE 10,,,100
TRANSFER .10,TYPEB,TYPEA
TYPEA QUEUE AQUEUE
  SEIZE INSPA
  DEPART AQUEUE
  ADVANCE 6,2
  RELEASE INSPA
  TRANSFER .10,REJA,ACCA
```

ACCA TERMINATE 1

REJA TERMINATE 1

```
TYPEB QUEUE BQUEUE
```

```
  SEIZE INSPB
```

```
  DEPART BQUEUE
```

```
  ADVANCE 10,2
```

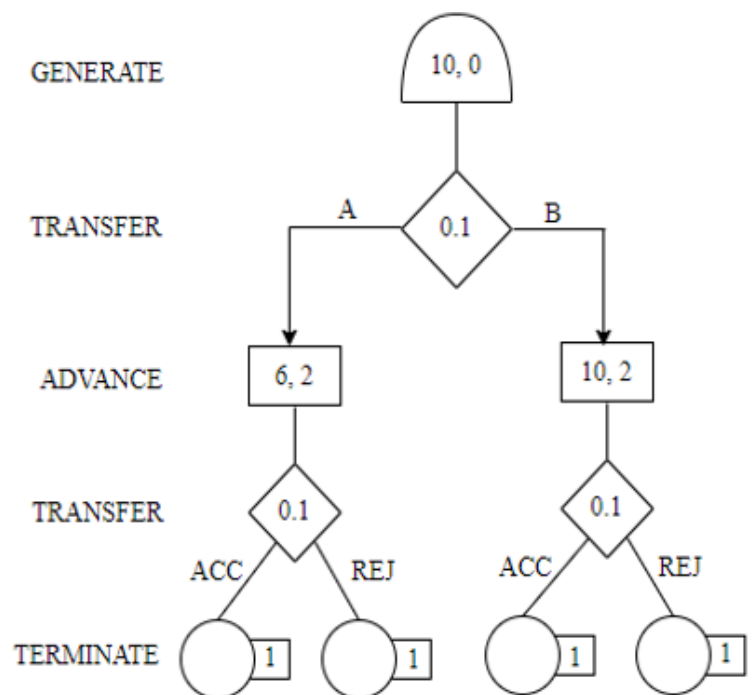
```
  RELEASE INSPB
```

```
  TRANSFER .10,REJB,ACCB
```

ACCB TERMINATE 1

REJB TERMINATE 1

START 100



File Edit Search View Command Window Help

Friday, June 19, 2026 18:54:20

NAME	VALUE
ACCA	9.000
ACCB	17.000
AQUEUE	10002.000
BQUEUE	10000.000
INSPA	10003.000
INSPB	10001.000
REJA	10.000
REJB	18.000
TYPEA	3.000
TYPEB	11.000

FACILITY	ENTRIES	UTIL.	AVE.	TIME	AVAIL.	OWNER	PEND	INTER	RETRY	DELAY
INSPB	89	0.871	9.871	1	0	0	0	0	0	0
INSPA	11	0.075	6.849	1	0	0	0	0	0	0

QUEUE	MAX	CONT.	ENTRY	ENTRY(0)	AVE.CONT.	AVE.TIME	AVE.(-0)	RETRY
BQUEUE	1	0	89	37	0.115	1.300	2.225	0
AQUEUE	1	0	11	11	0.000	0.000	0.000	0

Report is Complete.